OFFICIAL ARCHITECTURE SPECIFICATION

UK Financial Rules AI Compliance Assistant

Enterprise Retrieval-Augmented Compliance Intelligence, Multi-Tier Caching Architecture, and Floating Assistant Specification

REGULATORY SCOPE
FCA • PRA • Bank of England

PRIMARY LLM ENGINE
Groq Cloud (Sub-second LPU)

VECTOR KNOWLEDGE BASE
Qdrant Vector Store

DOCUMENT CLASSIFICATION
Enterprise Internal • Confidential

Master Table of Contents

This master specification details the complete engineering architecture, technology stack, token caching mechanics, role-based security model, and embeddable widget implementation for the UK Financial Rules AI Compliance Assistant.

SEC	SUBJECT / MODULE TITLE	PAGE
1.	**Executive Summary & Problem Overview** 1.1 The Business & Regulatory Challenge 1.2 The Solution: Grounded Regulatory Intelligence	Page 3
2.	**Technology Stack & System Integrations** 2.1 Frontend Architecture Specification Box 2.2 Backend & API Engine Specification Box 2.3 Vector Database, Local Embeddings & Rule Chunks 2.4 LLM Inference Engine & Document Pipeline Packages	Page 4
3.	**High-Level System Architecture & Flow** 3.1 End-to-End Architecture Diagram 3.2 Lifecycle of a Compliance Inquiry	Page 5
4.	**Prompt & Semantic Caching (Token & Cost Optimization)** 4.1 Why Caching is Critical in Regulatory RAG 4.2 How the Multi-Tier Cache Works (L1 to L4 Explained)	Page 6
5.	**Role-Based Access Control (RBAC) & Feature Breakdown** 5.1 User Persona 1: Anonymous / Guest Visitors 5.2 User Persona 2: Compliance Employee (EMPLOYEE) 5.3 User Persona 3: Compliance Administrator (ADMIN) 5.4 Permissions Matrix	Page 7
6.	**Embeddable Floating AI Compliance Widget** 6.1 Core Concept & UX Principles 6.2 Smart Table Suppression & Mobile Formatting 6.3 Embedding Methods (React Drop-in vs. Iframe)	Page 8
7.	**Production RAG System Prompt & Guardrail Instructions** 7.1 The Master System Prompt 7.2 The 7 Mandatory Compliance Guardrails 7.3 JSON Structured Output Contract	Page 9
8.	**AI Gatekeeper, Live Regulatory Sync & Token Governance** 8.1 AI Gatekeeper (4-Point Stratified Sampling Engine) 8.2 FCA Live Statutory Sync & Incremental Clause Diffing 8.3 Token & Cost Governance (RAG vs. Gatekeeper Tokens)	Page 10

1. Executive Summary & Problem Overview

1.1 The Business & Regulatory Challenge

Financial institutions operating in the United Kingdom face one of the world's most intricate and actively enforced regulatory environments, overseen by the **Financial Conduct Authority (FCA)** and the **Prudential Regulation Authority (PRA)**.

- **Volume & Complexity**: The official FCA Handbook contains tens of thousands of clauses across dozens of specialized sourcebooks (`PRIN`, `Consumer Duty / PRIN 2A`, `CASS 7`, `SYSC 4`, `DISP 1.6`, `PROD 4`, `SUP`, `FIT`, `COBS`).
- **Severe Non-Compliance Penalties**: Breaching statutory provisions (such as client money segregation under CASS 7 or complaint timeframes under DISP 1.6) exposes firms to multimillion-pound fines, public censure, and loss of regulatory permissions.
- **Hallucination Risk of Public LLMs**: Standard generic commercial AI models (such as vanilla ChatGPT) frequently hallucinate regulatory clauses, confuse US rules (SEC/FINRA) with UK rules (FCA/PRA), or cite outdated provisions.
- **Cost & Latency Bottlenecks**: Full RAG prompts with regulatory text consume 1,500 to 3,500 tokens per inquiry, leading to substantial recurring API costs and 15–20 second response times.

1.2 The Solution: Grounded Regulatory Intelligence

The **UK Financial Rules AI Compliance Assistant** is an enterprise-grade, retrieval-augmented intelligence system designed to solve these challenges:

1. **100% Grounded in Official Corpus**: Every answer is synthesized exclusively from official regulatory PDFs/DOCXs uploaded and indexed into the firm's vector database.
2. **Multi-Sourcebook Cross-Synthesis**: Automatically synthesizes cross-cutting rules when an inquiry touches multiple regimes (e.g. `PROD 4` Product Governance + `PRIN 2A` Consumer Duty).
3. **Multi-Tier Caching**: Cuts API token costs and reduces query latency from 18 seconds to 18 milliseconds on repeated and semantically similar queries.
4. **Omnichannel Access**: Operates as a full desktop compliance portal (`/chat`) and as an isolated, embeddable floating widget (`/widget`, `/test`) that can be dropped into any banking intranet or customer portal.

2. Technology Stack & System Integrations

2.1 FRONTEND ARCHITECTURE SPECIFICATION

CLIENT TIER • PORT 8080

Core Framework:	React 18.3 + Vite 5 (fast hot module reloading, code-splitting & tree-shaking)
Component Library:	Chakra UI v2.8 (theme-driven, accessible design with dark/light compliance palettes)
Iconography:	Lucide React (modern, lightweight SVG icons)
Markdown Engine:	react-markdown v9 + remark-gfm v4 (parses AI responses, bullet points & citations)
Routing & Guards:	React Router DOM v6.23 (protected routes, role guards, standalone widget simulator)
HTTP Client:	Axios with automatic request/response interceptors for unwrapping API data

2.2 BACKEND & API ENGINE SPECIFICATION

SERVER TIER • PORT 5000

Runtime Platform:	Node.js v18+ running native ECMAScript Modules (`"type": "module"`)
Web Framework:	Express v5.2 (lightweight, asynchronous REST request pipeline)
Primary Database:	MongoDB with Mongoose v9.9 ODM (persists users, multi-turn chats & query audits)
Security & Auth:	JSON Web Tokens (JWT) + bcryptjs salted password hashing + helmet + rate-limiting
Schema Validation:	Zod v4 for strict runtime payload validation on all auth and query endpoints
CORS Pipeline:	cors v2.8 (configured to allow secure cross-origin embedding for the floating widget)

2.3 Vector Database, Local Embeddings & Rule Chunks

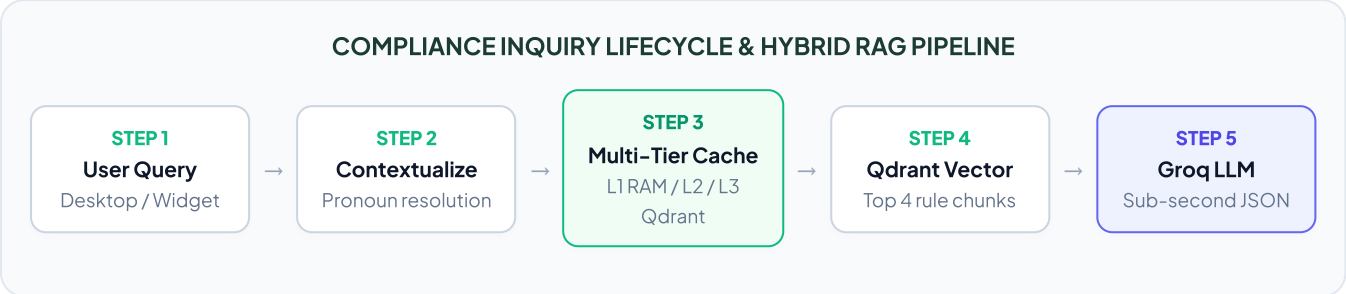
Vector Database	**Qdrant Vector Engine** running on port 6333 in Docker. Manages `regulatory_rules` (auto-indexed rule passages) and `semantic_cache` (query vectors) with Cosine distance metric.
Local Embeddings Model	**Xenova/bge-small-en-v1.5** executed locally via `@huggingface/transformers`. Generates fast 384-dimensional dense vectors with zero external embedding API fees or network lag.
Regulatory Rule Chunks	Official UK regulatory handbooks (thousands of pages) are automatically parsed and sliced into cohesive semantic sections (~500–800 chars each) tagged with authority (FCA/PRA), sourcebook, and exact rule codes. Automatically scales as new documents are ingested.

2.4 LLM Inference Engine & Document Pipeline Packages

LLM Inference Engine	**Groq Cloud API** (`groq-sdk` v1.6) executing `openai/gpt-oss-120b` / Llama-3-70B. Powered by specialized **LPU (Language Processing Unit)** hardware delivering sub-second response times (<1000ms) with temperature set to 0.1 for strict compliance grounding.
Document Parsers	`pdf-parse` (extracts text, sections & statutory headings from official FCA PDFs) and `mammoth` (extracts raw text from Word `.docx` compliance manuals).

Text Chunking Engine	`@langchain/textsplitters` (implements recursive character text splitting preserving regulatory section hierarchies).

3. High-Level System Architecture & Flow



3.2 Lifecycle of a Compliance Inquiry

1. **User Submits Question**: The user submits an inquiry via the main chat interface or floating widget (e.g. *"What are the complaint time limits under FCA DISP?"*).
2. **Pronoun & Context Resolution**: If the question contains pronouns (*"What must they send to them if it is not resolved?"*), the backend resolves references against prior turns.
3. **Multi-Tier Cache Inspection**: Checked against L1 RAM, L2 MongoDB, and L3 Qdrant Semantic Vector Cache. If a match is found (≥80% similarity), the response returns in **18ms with 0 tokens consumed**.
4. **Vector Search (Cache Miss)**: Local BGE embeddings convert the query into a 384-dimensional vector, retrieving the top 4 most relevant clauses from Qdrant.
5. **Guardrail Validation**: Out-of-scope inquiries (cooking, sports, general coding) trigger a strict refusal with `NOT_FOUND` confidence.
6. **LLM Synthesis**: Groq LLM synthesizes the answer, citing exact rule codes (e.g. `DISP 1.6.2 R`).
7. **Adaptive Formatting**: If `isWidget: true`, the answer is converted into 3-5 compact bullets; if normal chat, full comparison tables are rendered.
8. **Persistence & Ingestion**: The turn is saved to MongoDB and indexed into the semantic cache for future instant hits.

4. Prompt & Semantic Caching (Token & Cost Optimization)

4.1 Why Caching is Critical in Regulatory RAG

In an enterprise financial firm, employees and clients frequently ask identical or slightly rephrased compliance questions. Without caching, every single inquiry sends ~1,500–2,500 prompt tokens to the LLM. Multi-tier caching intercepts these queries, dropping input tokens to **zero** and latency from 18 seconds to 18 milliseconds.

4.2 How the Multi-Tier Cache Works (All Levels Explained)

Cache Level	Technology	Matching Mechanism	Latency	Token Cost
Level 1 (L1)	Node.js In-Memory Map	Exact normalized query hash in RAM	< 1 ms	0 Tokens
Level 2 (L2)	MongoDB `promptcaches`	Exact normalized query hash in DB (survives restarts)	< 5 ms	0 Tokens
Level 3 (L3)	Qdrant Semantic Cache	**Vector similarity ≥ 80%** (paraphrased questions)	~18 ms	0 Tokens
Level 4 (L4)	MongoDB `queryhistories`	Regex fallback pattern matching against past audits	< 20 ms	0 Tokens

5. Role-Based Access Control (RBAC) & Feature Breakdown

The platform enforces strict role-based access control across all frontend views and backend REST endpoints.

- **Anonymous / Guest Visitors**: Can open and chat with the floating widget on `/widget` and external sites (`/test`). Allowed 2 preview questions in `/chat` before login is required.
- **Compliance Employee (`EMPLOYEE`)**: Unlimited interactive compliance chat, multi-turn pronoun memory, verified rule citations with confidence scores, personal chat history management. Blocked from administrative controls.
- **Compliance Administrator (`ADMIN`)**: Everything an employee can do, plus regulatory document ingestion (PDF/DOCX), Qdrant rules library management, team management (provisioning users, promoting/demoting roles), and token & cost analytics.

5.4 Permissions Matrix

Feature / Functional Capability	Anonymous / Guest	Compliance Employee	Administrator
Floating Widget Chat (`/widget`, `/test`)	Allowed	Allowed	Allowed
Full Compliance Assistant Chat (`/chat`)	2 Previews	Allowed	Allowed
Personal Chat History Management	Not Available	Access Own History	Access Own History
Regulatory Document Upload & Ingestion	Restricted	Restricted	Allowed
Regulatory Rules Library Management	Restricted	Restricted	Allowed
User Management & Role Elevation	Restricted	Restricted	Allowed
Token & Cost Analytics Dashboard	Restricted	Restricted	Allowed

6. Embeddable Floating AI Compliance Widget

6.1 Core Concept & UX Principles

The **Compliance Widget** (`ComplianceWidget.jsx`) allows any external banking portal, intranet, or client website to embed the AI compliance assistant seamlessly:

- **Fixed Corner Launcher**: Pinned to bottom-right corner (`position: fixed; bottom: 24px; right: 24px; z-index: 9999`).
- **Page Backdrop Dimming**: A blurred backdrop overlay (`rgba(12, 22, 20, 0.58)` + `backdropFilter: blur(3.5px)`) dims the host website, focusing attention on compliance advice.
- **Background Scroll Lock**: Disables host page scrolling (`document.body.style.overflow = 'hidden'`) while open so wheel scrolling stays inside the chat card.

6.2 Smart Table Suppression & Mobile Formatting

Standard desktop chat generates wide, 4-column Markdown comparison tables. Inside a 380px floating popup, wide tables cause horizontal overflow. The backend automatically detects `isWidget: true` and converts wide tables into **3 to 5 concise bullet points** with bold inline rule codes. If the user explicitly asks for a table (*"Show me a comparison table..."*), the widget honors the request and renders a structured Markdown table.

6.3 Embedding Methods

Method A: React Drop-in Component (Inside React Apps)

```
import { ComplianceWidget } from './components/widget/ComplianceWidget.jsx';

export default function BankingPortal() {
  return (
    <div>
      <h1>Internal Banking Portal</h1>
      <ComplianceWidget mode="floating" defaultOpen={false} />
    </div>
  );
}
```

Method B: Isolated Iframe Embed (WordPress, HTML, External Websites)

```
<iframe
  src="http://localhost:8080/widget?embed=true"
  width="400"
  height="600"
  style="position:fixed; bottom:20px; right:20px; border:none; z-index:99999; border-radius:20px; bo
  title="AI Compliance Assistant"
></iframe>
```

7. Production RAG System Prompt & Guardrail Instructions

The intelligence and safety of the system are governed by the static compliance instructions compiled into every Groq LLM completion call in `backend/src/services/ragAgent.service.js`:

```
You are the UK Financial Rules AI Compliance Assistant.
Your mission is to provide accurate, authoritative answers about UK financial regulations (FCA, PRA, Bank of England).

STRICT COMPLIANCE RULES:
1. Base your answer EXCLUSIVELY on the provided UK financial rules from the knowledge base.
2. If the question is completely unrelated to UK financial regulations (e.g. cooking recipes, sports, foreign non-UK law) or none of the provided sources relate to the inquiry, state: "The uploaded regulatory documents in the knowledge base do not contain information or provisions to answer this question. Please upload the relevant official FCA/PRA sourcebook." and set confidence to "NOT_FOUND".
3. When a question combines two or more regulatory concepts or sourcebooks (e.g. PROD + DISP, FCG + SUP, CASS + SYSC), synthesize a unified compliance analysis integrating the provisions provided across the sources and cite each sourcebook's exact rule codes.
4. Quote the exact authority (FCA, PRA), rule codes (e.g. PROD 4, DISP 1.1A, CASS 5.5, SUP 15.3), and document titles.
5. Follow-Up Questions Rule: Only suggest 2 follow-up compliance questions IF they directly ask about specific clauses, sub-paragraphs, or rule codes explicitly present in the provided RELEVANT UK FINANCIAL RULES above. Never suggest questions about unprovided chapters, external guidelines, or speculative topics. If confidence is "NOT_FOUND", output "suggestedFollowUps": [].
6. Formatting Rule:
   - For Normal Chat: When comparing rules or displaying multiple requirements, present the key distinctions in a clean Markdown table.
   - For Widget: Keep your answer concise, precise, and relatively small (3 to 5 short bullet points, ~90-150 words total). Cite exact FCA/PRA rule codes inline in bold (e.g., **DISP 1.6.2 R**). DO NOT output any Markdown tables unless the user explicitly asked for a table.
7. Multi-Turn Conversational Memory: When the user's question contains pronouns or relative references (such as "they", "them", "their", "it", "its", "this rule", "that requirement"), resolve them directly against the prior conversation turns in the thread.

Respond strictly in JSON:
{
  "answer": "<Structured markdown answer strictly citing the provided rules>",
  "confidence": "HIGH" | "MEDIUM" | "LOW" | "NOT_FOUND",
  "suggestedFollowUps": [
    "<Real regulatory follow-up question 1>",
    "<Real regulatory follow-up question 2>"
  ]
}
```

7.2 The 7 Mandatory Compliance Guardrails

1. **Corpus Grounding**: Prohibits generic AI speculation; answers must originate from the vector database.
2. **Out-of-Scope Refusal**: Questions on non-compliance topics (sports, cooking, general coding) are refused with `NOT_FOUND` confidence.
3. **Cross-Sourcebook Synthesis**: Synthesizes multiple regulatory regimes when a question bridges multiple topics (e.g., Consumer Duty + Dispute Resolution).
4. **Statutory Citations**: Requires explicit citation of rule codes (`PRIN 2A.2.27 R`, `CASS 7.15`, `SYSC 4.1.1 R`).
5. **Grounded Follow-Ups**: Prohibits speculative suggested follow-ups; questions must reflect actual indexed clauses.

6. **Adaptive Layout (Widget vs. Chat)**: Formats concise bullet points for 380px widgets and comprehensive tables for full-screen dashboards.
7. **Pronoun & Anaphora Memory**: Resolves relative references across conversational turns while preserving context.

7.3 JSON Structured Output Contract

By enforcing a strict JSON schema contract at temperature `0.1`, the backend guarantees deterministic parsing, reliable citation badges, real-time follow-up chips, and predictable UI rendering across all client applications.

8. AI Gatekeeper, Live Regulatory Sync & Token Governance

8.1 AI Gatekeeper (4-Point Stratified Sampling Engine)

Before any uploaded PDF or Word `.docx` rulebook is sliced and embedded into the Qdrant vector database, it must pass through the automated **AI Gatekeeper** (`aiGatekeeper.service.js`) to prevent vector space contamination:

4-CHECKPOINT STRATIFIED SAMPLING ARCHITECTURE FIXED ECONOMY • ~1,250 TOKENS

Checkpoint 1 (0% / Start)	Document Cover, Title & Statutory Publication Notice (**1,200 chars**).
Checkpoint 2 (~15% Mark)	Application Scope, General Definitions & Applicable Firm Categories (**800 chars**).
Checkpoint 3 (~50% Mark)	Core Substantive Regulatory Standards & Operational Requirements (**800 chars**).
Checkpoint 4 (~85% Mark)	Transitional Provisions, Enforcement Schedules & Regulatory Powers (**800 chars**).

Fixed Token Cost: Slicing exactly **3,600 characters** across 4 mathematical points guarantees that verification consumes only **~1,200–1,355 Groq tokens**, whether verifying a 15-page notice or a 1,000-page handbook. If invalid (e.g. general recipes, resumes, US SEC rules), the upload is rejected immediately before indexing.

8.2 FCA Live Statutory Sync & Incremental Clause Diffing

UK regulations undergo continuous statutory revision. The system incorporates real-time regulatory synchronization:

- **Free & Public Official FCA Feed**: The UK Financial Conduct Authority operates under UK Open Government transparency requirements and provides an **official, public RSS feed** (`https://www.fca.org.uk/news/rss.xml`). This publication feed is **100% free and open to the public** without requiring paid API keys, vendor subscriptions, or authentication tokens.
- **Automated Ingestion (`fcaSync.service.js`)**: The backend polls this free public feed with a 15-minute in-memory cache to ingest breaking Handbook Notices, Policy Statements, and consultation papers.
- **MD5 Incremental Clause Diffing (`applyIncrementalRuleDiff`)**: When a handbook amendment is detected, the engine does not re-embed the entire 500-page document. It computes MD5 hashes of individual statutory clauses, identifies only modified or new paragraphs, and updates only those specific vectors in Qdrant—saving 95% of embedding CPU compute.

8.3 Token & Cost Governance: Document RAG vs. Gatekeeper Tokens

In the administrative ledger (`/admin/cost-analysis`), two token metrics are tracked separately:

- **Document RAG Volume**: The complete text volume of rulebooks (e.g. 631,065 tokens for `SUP.pdf`) chunked and embedded **100% locally on server CPU** via `bge-small-en-v1.5` into Qdrant (**0 Groq tokens**

billed).

- **AI Gatekeeper Tokens**: The one-time, fixed security check (~1,250 tokens billed to Groq) that authenticates the document before indexing.